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RESEARCH ARTICLE

MEDICAL PHYSICS

STimulated emission depletion (STED) magnetic particle imaging

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Abstract

Background: Magnetic particle imaging (MPI) is an in vivo method for detecting magnetic nanoparticles for cell tracking and molecular target imaging. Recently developed human-sized MPI scanners allow a spatial resolution of only 5–10 mm. High-resolution MPI is required to precisely locate the magnetic nanoparticles in living organisms.

Purpose: This study proposed a high-resolution imaging method by introducing the STimulated Emission Depletion (STED) fluorescence microscopy principle via a donut-shaped point spread function (PSF).

Methods: We searched for the donut focal spot by adding a direct current offset stimulation magnetic field in the direction of the excitation field and receiver coil in MPI. When the offset stimulation field is greater than the excitation field amplitude, a donut-shaped focal spot can be formed and the donut radius increases with the offset field amplitude. The formation of the donut focal spot was theoretically evaluated, and the center signal of the donut focal spot was depleted with a strict explanation. The point-source phantom was experimentally imaged on a field-free-point-based MPI scanner and exhibited a donut-shaped line profile under a large offset stimulation field. The phantom images were simulated on a field-free-line-based MPI scanner to generate donut-shaped PSFs and images.

Results: By applying an optimized STED factor in the subtraction of the donut PSF from the regular Gaussian PSF, a much smaller PSF could be obtained for deconvolution-based image reconstruction. The reconstructed images exhibited sub-millimeter resolution, small root mean square error, high peak signal-to-noise ratio, and high structural similarity index.

Conclusion: The donut-shaped PSF obtained using the STED imaging method can be used to improve the MPI resolution by breaking the Langevin magnetization barrier.

KEYWORDS
donut-shaped focal spot, field-free line, magnetic particle imaging, point spread function, stimulated emission depletion

1 | INTRODUCTION

Magnetic particle imaging (MPI) is a novel in vivo method for the sensitive detection of magnetic nanoparticles without ionizing radiation for T-cell tracking,[1] stem cell tracking,[2] blood vessel imaging,[3–5] and molecular target imaging.[6,7] To date, only small animal MPI scanners are commercially available with a low resolution of 1–6 mm.[8] Recently developed human-sized MPI scanners allow a spatial resolution of only 5–10 mm.[9] High-resolution MPI is required to precisely locate magnetic nanoparticles in living organisms.[10]

Several techniques have been developed for high-resolution MPI. For example, superconducting coils were used to improve the selection field gradient,[11] and pulsed excitation was used to overcome the relaxation barrier of large-sized magnetic particles.[12] Methods to improve resolution also include calibrated reconstruction and artificial intelligence (AI) algorithms.[13] For example, graphics processing unit (GPU)-accelerated parallel image reconstruction algorithms have been used to improve image resolution.[14] Machine learning algorithms have been applied to improve the resolution of mouse islet transplantation models.[15] All these methods are based on the fundamental Langevin magnetization physics of magnetic nanoparticles, such that the fundamental Gaussian or Lorentzian point spread function (PSF) has not been changed to break the resolution barrier.[16]

This study proposed a high-resolution MPI method based on the STimulated Emission Depletion (STED) fluorescence microscopic imaging principle.[17] The basic theory of STED was proposed by Professor Hell in 1994, who introduced a small donut-based PSF in addition to the regular Gaussian PSF.[18] The STED method surpasses the traditional optical diffraction limit and achieves a much higher resolution than traditional fluorescence imaging.[19] In 2000, Professor Hell experimentally obtained a central dark donut focal spot using the STED method and achieved a 3-fold resolution improvement.[20] In 2007, Professor Hell summarized STED fluorescence microscopy based on donut focal spots, which spanned from the "micron" microscopic era to the "nano" era.[21] MINSTED and MINFLUX technologies were recently proposed by changing the radius and position of the donut focal spot to accurately locate fluorescent spots and improve resolution.[22,23]

The key point of STED high-resolution imaging is the finding of a small donut-shaped focal spot in MPI. Recently, a donut-shaped signal pattern was proposed on a non-field-free-point-based MPI scanner, in which a gradient descent method was used to find a valley point for localization.[24] This donut-shaped signal pattern has a large radius and may not be helpful in improving the resolution. In this study, we searched for a donut focal spot based on an additional offset stimulation magnetic field in the direction of the excitation field and receiver.[25] As the offset field increases to nearly half of the excitation field amplitude, a critical point is formed with a minimal 3rd harmonic signal owing to the jump-like behavior of the phase,[26] which is highly sensitive to the physical properties of the magnetic nanoparticles and the surrounding environment. This critical offset field may indicate a donut-shaped focal spot along the orthogonal selection field direction[27] and is worth further exploration (Figure 1a).

In this study, a systematic search of a donut-shaped focal spot was conducted by changing the collinear offset magnetic field. Under different stimulation offset conditions, the size of the focal spot was investigated using theoretical deduction, imaging simulation, and scanner experiments. A focal spot modulation technique and deconvolution algorithm (Figure 1b) were developed for image reconstruction.

2 | THEORY

2.1 | STED-MPI signal

In STED-MPI, we combine a cosine excitation field with a collinear stimulation DC magnetic field along the field-free line (FFL). The imaging space is covered by two orthogonal gradient fields in a typical FFL-based MPI scanner (Figure 2). We assume that the alternating excitation field has a frequency $\omega_0 = 2\pi/T$ and an amplitude H_{AC}. The stimulation field was defined as an offset direct-current field H_{DC} in the z-direction. The selection gradient fields G forming the FFL are in the x-y plane. The total magnetic field is

$$\boldsymbol{H}(x, y, z, t) = \begin{pmatrix} -Gx \\ Gy \\ H_{DC} - H_{AC}\cos\omega_0 t \end{pmatrix}. \quad (1)$$

Using a pair of detection coils in the z direction, the STED signal can be expressed as

$$s_z(t) = -\frac{dM_z(t)}{dt} = -\frac{\partial M_z}{\partial H_z}\frac{dH_z}{dt}, \quad (2)$$

which is proportional to the time variation of the z-component of the magnetization.

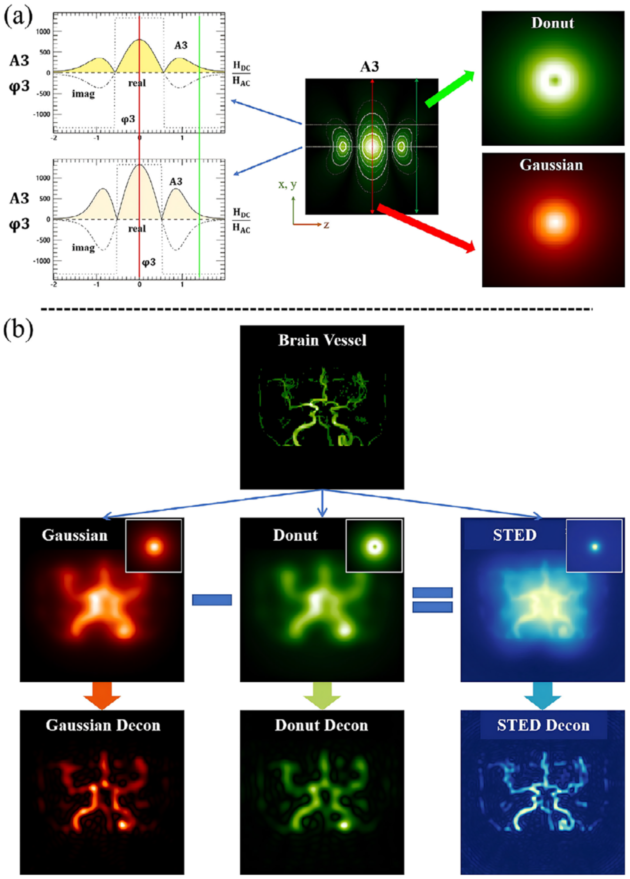

FIGURE 1 Illustration of the significance of the donut-shaped PSF in STED-MPI. (a) The 3rd Harmonic STED signal component exhibits different focal spot shapes depending on the DC field amplitude, for example, a Gaussian bright focal spot when $H_{DC} = 0$ and a donut focal spot with depletion in the center when $H_{DC} > H_{AC}$. (b) A deconvolution algorithm based on Gaussian, Donut, and STED PSF was applied to generate the reconstructed images. MPI, magnetic particle imaging; PSF, point spread function; STED, stimulated emission depletion.

The stimulated emission signal frequency components are defined as

$$S_{n,z} = \frac{1}{T}\int_{-T/2}^{T/2} s_z(t) e^{-in\omega_0 t} dt = -\frac{1}{T}\int_{-T/2}^{T/2} \frac{\partial M_z}{\partial H_z}\frac{dH_z}{dt} e^{-in\omega_0 t} dt. \tag{3}$$

The stimulated emission signal frequency components can be expressed as an integral versus the excitation and stimulation magnetic fields,

$$S_{n,z} = -\frac{1}{T}\int_{H_z(-T/2)}^{H_z(T/2)} \frac{\partial M_z}{\partial H_z} e^{-in\omega_0 t(H_z)} dH_z. \tag{4}$$

Based on the cosine function of the excitation field, time can be expressed as

$$t(H_z) = \begin{cases} -\frac{1}{\omega_0}\arccos\frac{H_z - H_{DC}}{H_{AC}}, & -\frac{T}{2} \le t \le 0 \\ \frac{1}{\omega_0}\arccos\frac{H_z - H_{DC}}{H_{AC}} & 0 \le t \le \frac{T}{2} \end{cases}. \tag{5}$$

The nth harmonic signal can be expressed as

$$S_{n,z} = -\frac{1}{T}\int \frac{\partial M_z}{\partial H_z}\left(e^{in\arccos\frac{H_z - H_{DC}}{H_{AC}}} - e^{-in\arccos\frac{H_z - H_{DC}}{H_{AC}}} \right) dH_z. \tag{6}$$

The real part is zero when relaxation effects are ignored, thus the nth harmonic signal only includes the imaginary part

$$S_{n,z} = \frac{-2i}{T}\int \frac{\partial M_z}{\partial H_z}\sin\left(n\arccos\frac{H_z - H_{DC}}{H_{AC}}\right) dH_z. \tag{7}$$

Functions of the type $\sin(n\arccos\frac{H_z - H_{DC}}{H_{AC}})$ are similar to the Chebyshev polynomials of the second kind.[28]

According to the convolution formula, the nth harmonic signal can be written as a convolution involving only the z-direction magnetic field H_z, that is

$$S_{n,z}(H_{DC}) = \frac{-2i}{T}\frac{\partial M_z}{\partial H_z} * \left(U_{n-1}\left(\frac{H_{DC}}{H_{AC}}\right)\sqrt{1 - \left(\frac{H_{DC}}{H_{AC}}\right)^2} \right). \tag{8}$$

For Langevin magnetic nanoparticles, the spatial dependency of the nth harmonic signal picked up in the z direction ($S_{n,z}$) becomes

$$S_{n,z} = \frac{-2iM_0}{T}\left[\beta\mathcal{L}'(\beta|H|)\frac{H_{DC}^2}{|H|^2} + \mathcal{L}(\beta|H|)\frac{|H|^2 - H_{DC}^2}{|H|^3} \right] * \left(U_{n-1}\left(\frac{H_{DC}}{H_{AC}}\right)\sqrt{1 - \left(\frac{H_{DC}}{H_{AC}}\right)^2} \right). \tag{9}$$

The field amplitude of the magnetic field for $H_z = H_{DC}$ is

$$|H| = \sqrt{(Gr)^2 + H_{DC}^2}, \tag{10}$$

where $r = \sqrt{x^2 + y^2}$ is the distance of the point to the original point in the x-y plane, M_0 the saturation magnetization, and β particle-related parameter ($\frac{\mu_0 m}{k_B T^P}$).

The nth harmonic signal is the derivative of the magnetization, convolved with a set of Chebyshev polynomials. The convolution kernel part

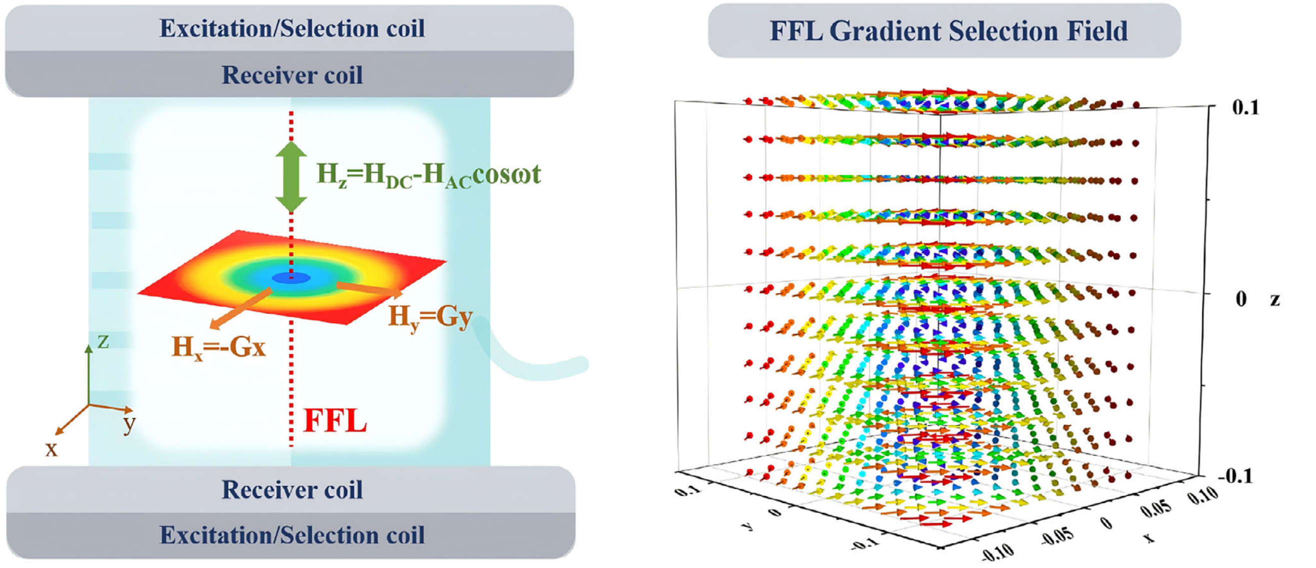

FIGURE 2 Schematic of STED-MPI. (Left) Magnetic nanoparticle excitation field, stimulation field, FFL selection field, and signal receiver in the STED-MPI system. (Right) 3D plot of the FFL selection gradient field with $G = 2.5$ mT/mm. FFL, field-free line; MPI, magnetic particle imaging; STED, stimulated emission depletion.

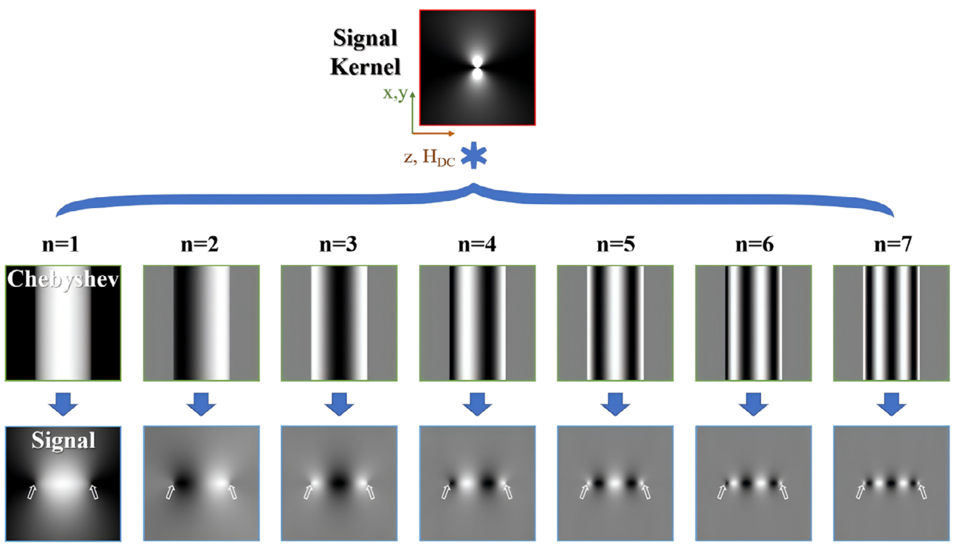

FIGURE 3 The harmonic signal components for Langevin particles under cosine excitation with a collinear offset field as stimulation can be calculated by the convolution of the base signal and Chebyshev kernels. Donut-shaped focal spots may be formed when the offset field is greater than the excitation field (indicted by hollow arrows).

$U_{n-1}\left(\frac{H_{DC}}{H_{AC}}\right)\sqrt{1-\left(\frac{H_{DC}}{H_{AC}}\right)^2}$ is zero when $H_{DC} > H_{AC}$ and $r = 0$ (Figure 3). The harmonic signal increases and then decreases as r increases along the vertical x-y direction, thus forming a donut-shaped PSF.

2.2 | Non-adiabatic STED signal

The relaxation effects of the Langevin particles may induce magnetization and signal lag during excitation.[29]

The STED signal of Debye particles induced in the receiver coils is given by[30]

$$s_{z\_Debye}(t) = -\frac{d}{dt}\left(M_z(t) * r_{Debye}(t)\right), \tag{11}$$

where $r_{Debye}(t)$ is the Debye relaxation kernel of magnetic particles and is defined as follows,

$$r_{Debye}(t) = \frac{1}{\tau_B} e^{-\frac{t}{\tau_B}} u(t), \tag{12}$$

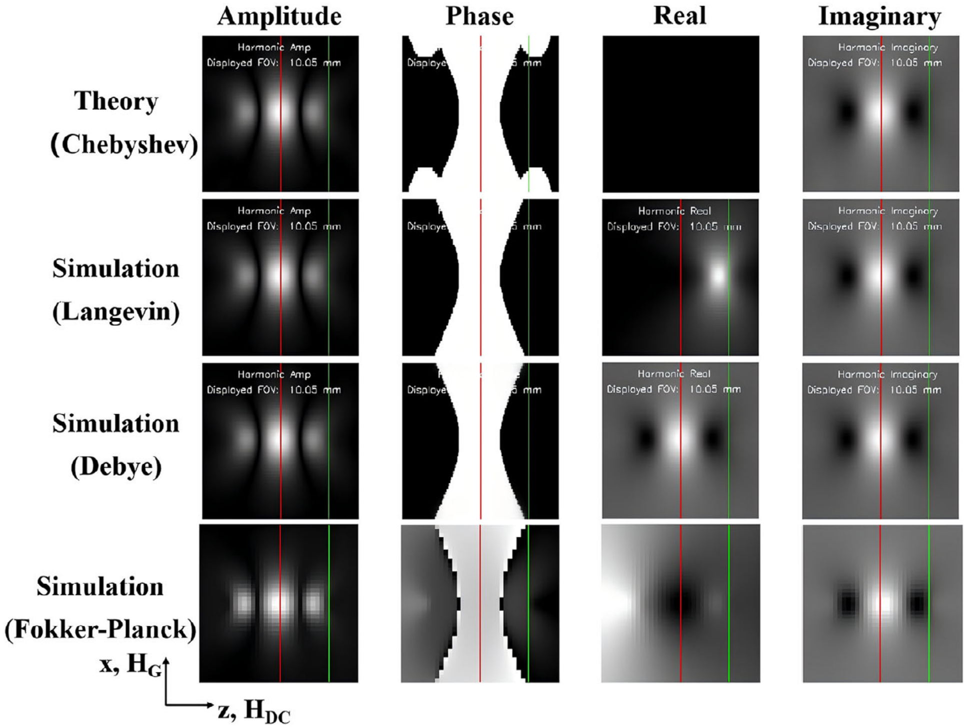

FIGURE 4 Signal frequency component A3 from theory and simulation. The red lines indicate a zero DC offset ($H_{DC} = 0$), along which the signal line profile exhibits a Gaussian-shaped PSF induced by the *x*-direction selection field. The green lines indicate $H_{DC} > H_{AC}$, showing a donut-shaped line profile induced by the *x*-direction selection field and *z*-direction DC offset field. PSF, point spread function.

where $u(t)$ is the Heaviside step function stepping at $t = 0$. The relaxation time τ_B can be approximated as the characteristic zero-field Brownian relaxation time[31]

$$\tau_B \approx \frac{3\eta V_h}{k_B T^P}, \tag{13}$$

where η represents the dynamic viscosity and V_h represents the hydrodynamic volume of the magnetic nanoparticles. After Fourier transform, the *n*th harmonic signal may be computed as

$$S_{n,z\_Debye}(r, H_{DC}) = S_{n,z}(r, H_{DC}) \cdot R_{n,Debye}(r, H_{DC}). \tag{14}$$

The relaxation frequency components $R_{n,Debye}\,(r, H_{DC})$ are defined as

$$R_{n,Debye}(r, H_{DC}) = \frac{\omega_0}{2\pi}\frac{e^{-i\theta_{n,\tau_B}}}{\sqrt{1 + (n\omega_0\tau_B)^2}}, \tag{15}$$

where θ_{n,τ_B} is the delay phase due to the relaxation effects,

$$\tan\left(\theta_{n,\tau_B}\right) = n\omega_0\tau_B. \tag{16}$$

The 3rd harmonic components based on the Langevin function with and without Debye relaxation are illustrated in Figure 4. Using a computational simulation framework[32] based on the Fokker–Planck equation,[33] the 3rd harmonic components were calculated by inputting the gradient fields and DC offsets. Donut-shaped PSFs are formed in all these simulation results when the DC offset is greater than the excitation field ($H_{DC} > H_{AC}$).The Langevin function and the Debye approximations are computationally efficient; however, the Weizenecker Fokker–Planck formalism, which incorporates both Brownian and Néel relaxation, may provide a more accurate estimation. The non-zero relaxation time[34] causes a phase delay in the signal[29] and affects the real and imaginary parts of the harmonic components. The more accurate models, mainly change the phase and imaginary part of the 3rd harmonic signal, which seems related to the nonadiabatic relaxation process and Debye approximation. To compensate for these changes and reduce the effect of relaxation on harmonic signals, we suggest using magnetic particles with smaller diameters to alleviate the effect of relaxation times.

3 | MATERIALS AND METHODS

3.1 | Donut PSF experiment

To evaluate the formation conditions of the donut focal spot, a point source was placed at the center of an field-

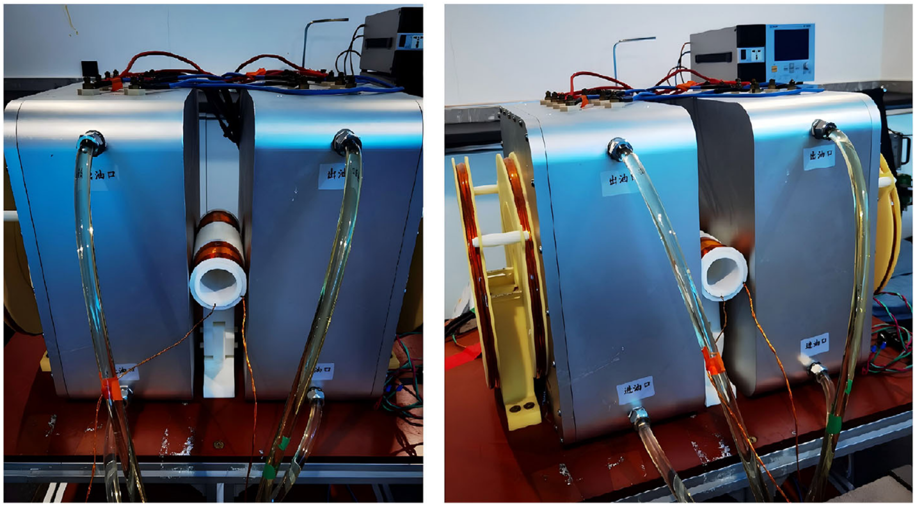

FIGURE 5 Donut-shaped PSF was experimentally assessed on an FFP-based MPI scanner with pictures from the front view (left) and side view (right). FFP, field-free-point; MPI, magnetic particle imaging; PSF, point spread function.

free-point (FFP)-based MPI scanner (Jiayu Technology Co., LTD, Liaoning, China, Figure 5). The sample source consisted of Resovist (Bayer AG, Germany) with a content of 40 μL and a concentration of 27.4 μg Fe/μL. The selection field was 0.85 mT/mm, and the excitation fields were 3–7 mT at a frequency of 20 kHz. The field of view (FOV) was 40 mm × 40 mm. The 3rd harmonic component was extracted using a narrowband MPI system. The line profiles along the direction orthogonal to the excitation and stimulation fields were evaluated to verify the formation of a donut focal spot. To assess the quality of a donut-shaped focal spot, the donut radius was defined as the distance between the donut center and the pixel with the maximum signal intensity. The donut thickness was defined as the difference between the central pixel signal intensity and the maximal signal intensity along the line profile.

3.2 | STED-MPI scanner imaging simulation

STED-MPI can be implemented on an FFL-based MPI scanner system, which includes an FFL selection field, collinear excitation and offset stimulation fields, and collinear signal acquisition for a imaging process by shifting and rotating FFL. Receiver coils can collect either Gaussian or donut-shaped PSF signals from magnetic nanoparticles for high-resolution imaging purposes. The STED harmonic signal and imaging simulations were performed using interactive data language (IDL). The IDL codes are available at https://github.com/GuangJia76/AIMIS3D-STED-MPI. The excitation magnetic field was set between 2 and 50 mT, and the excitation frequency was within 100 kHz. The sampling rate was set to 1 MHz. The DC offset stimulation magnetic field was set between -200% and 200% of the excitation magnetic field amplitude. Two DC offset magnetic fields were selected to generate Gaussian and donut focal spots. The phantom images could be uploaded for STED imaging and deconvolution-based reconstruction.[35]

In the image reconstruction simulation, the Gaussian focal spot and donut focal spot were subtracted to generate a smaller PSF (Figure 6). An STED factor was used during subtraction of two focal spots

$$PSF_{STED} = PSF_{Gaussian} - f_{STED} \cdot PSF_{Donut}. \qquad (17)$$

If the STED factor is too small, PSF_{STED} may have a similar FWHM as $PSF_{Gaussian}$, which may not efficiently improve the image resolution. If the STED factor is too large, negative signals will appear on and cause signal cancelation for objects at close distances, which introduces artifacts and hinders image reconstruction. According to the difference of Gaussian (DoG) theory,[36] PSF_{STED} can be close to a Gaussian shape with a narrow width if the STED factor is appropriately chosen. In this study, a 0% DC offset was set as the Gaussian PSF. A 105% DC offset was selected as a donut PSF and an STED factor of 1.9 was chosen for STED PSF generation.

The phantom images included multiple point sources, a 2D bar pattern phantom, bars with four different concentrations, a vascular phantom, a neoplasm phantom,[31] an american college of radiology (ACR) mammography phantom, a Shepp–Logan phantom, and brain vessel projection maps from time-of-flight magnetic resonance angiography (TOF-MRA) images.[37] Measured signal images were generated based on the convolution of a phantom image with Gaussian and donut PSFs, after which noise was added to mimic the electronic signal collection process. The Richardson–Lucy deconvolution algorithm[38] was used to generate higher-

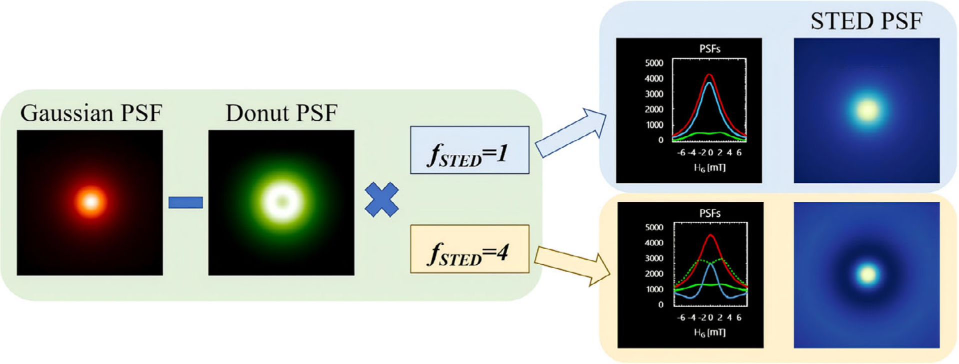

FIGURE 6 An STED factor can be used to generate a much smaller PSF than the bright Gaussian PSF during the focal modulation process. The red solid, green solid, green dashed, and blue solid curves represent the Gaussian, donut, donut times STED factor, and STED PSFs, respectively. PSF, point spread function; STED, stimulated emission depletion.

resolution images (Figure 1b). The reconstructed image quality was evaluated using the root mean square error (RMSE), peak signal-to-noise ratio (PSNR), and structural similarity index (SSIM).[39] RSME is defined as

$$RMSE = \sqrt{\frac{1}{N}\left(\sum_{i=1}^{N}\left(P_i - \hat{P}_i\right)^2\right)}, \tag{18}$$

where N represents the pixel number of the image, and P_i and $\hat{P}_i$ are the true and reconstructed pixel signal intensities, respectively. PSNR is expressed as

$$PSNR = 10\log_{10}\frac{MAX_P^2}{RMSE^2}, \tag{19}$$

where MAX_P is the maximum pixel signal intensity of the true phantom image. SSIM can be calculated from

$$SSIM = \frac{\left(2\mu_P\mu_{\hat{P}} + c_1\right)\left(2\sigma_{P\hat{P}} + c_2\right)}{\left(\mu_P^2 + \mu_{\hat{P}}^2 + c_1\right)\left(\sigma_P^2 + \sigma_{\hat{P}}^2 + c_2\right)}, \tag{20}$$

where μ_P and $\mu_{\hat{P}}$ are the means of the true and reconstructed phantom images, respectively; σ_P and $\sigma_{\hat{P}}$ are the variances of the true and reconstructed phantom images, respectively; $\sigma_{P\hat{P}}$ is the covariance between the images; and c_1 and c_2 are two constants.[40]

4 | RESULTS

4.1 | STED signal simulation results

A regular Gaussian focal spot was formed when there was no offset magnetic field ($H_{DC} = 0$). By increasing the offset magnetic field H_{DC} ($H_{DC} > H_{AC}$), the stimulated emission signal was depleted at the center of the donut focal spot. The stimulated emission signal increased and then decreased in the orthogonal gradient H_G direction, forming a donut focal spot (Figure 1a).

By varying the excitation and offset stimulation magnetic field, either Gaussian or donut PSFs were formed along the orthogonal gradient-selection field directions. When $H_{DC} < H_{AC}$ and $H_{DC} \neq 0.5 * H_{AC}$, Gaussian PSFs were formed and exhibited a relatively stable full width at half maximum (FWHM) (Figure 7a) and a large variation in the signal (Figure 7b). The donut PSFs showed a large radius (Figure 7c) and small thickness (Figure 7d), when $H_{DC} = 0.5 * H_{AC}$. The radius and thickness were small when $H_{DC} > H_{AC}$ and increased with a greater H_{DC}.

4.2 | FFP-based scanner experimental results

The FFP was not symmetric because of the DC offset field parallel to the AC excitation field. We used FFP to evaluate harmonic signals at different DC offsets, which is similar to the finding of the critical point when H_{DC} was one-half of H_{AC}.[41] We observed donut-shaped line profile when H_{DC} was greater than H_{AC}.

Donut-shaped line profiles were plotted from the 3rd harmonic signal natural logarithm maps, using different excitation field amplitudes on an FFP-based MPI scanner when the offset field was greater than 100% of the excitation field strength (Figure 8). The signals of the donut line profiles were much smaller than those of the Gaussian line profiles with a zero offset stimulation field.

4.3 | STED-MPI simulation results

The reconstructed phantom images using the STED-MPI method and deconvolution algorithm exhibited higher quality than Gaussian-based or donut-based

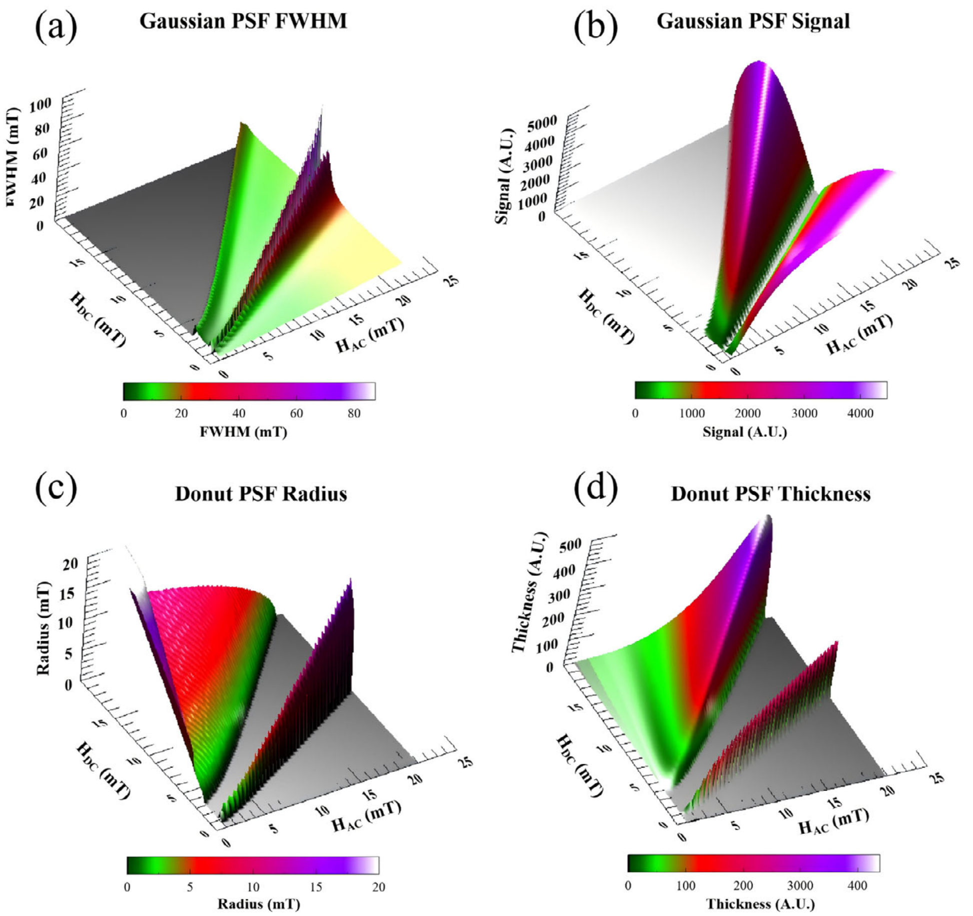

FIGURE 7 Surface plot of the FWHM of the Gaussian PSFs at different excitation and offset stimulation magnetic fields. Gaussian PSFs were formed with a relatively stable FWHM when $H_{DC} < H_{AC}$ and $H_{DC} \neq 0.5 * H_{AC}$ (a). Surface plot of the signal of the Gaussian PSFs at different excitation and offset stimulation magnetic fields. Gaussian PSFs exhibited a decreasing signal with a greater H_{DC} when $H_{DC} < 0.5 * H_{AC}$ (b). Surface plot of the radius of the donut PSFs at different excitation and offset magnetic fields. A large donut PSF was formed when $H_{DC} = 0.5 * H_{AC}$. A significantly smaller donut PSF was formed when $H_{DC} > H_{AC}$ (c). Surface plot of the relative signal or thickness of the donut PSFs at different excitation and offset magnetic fields. The donut PSFs exhibited a small signal owing to the large offset stimulation field $H_{DC} > H_{AC}$ (d). PSF, point spread function.

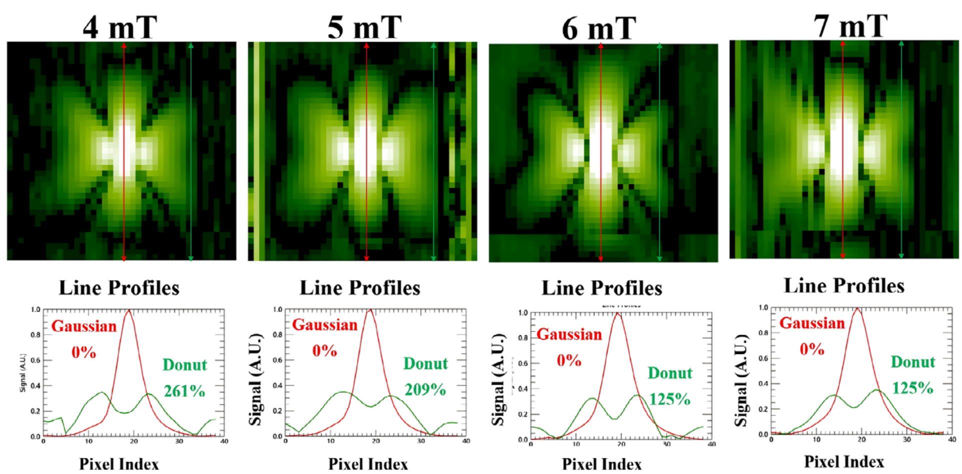

FIGURE 8 3rd harmonic signal maps and line profiles of a point source phantom from an FFP-based MPI scanner using different excitation field amplitudes. Gaussian-shaped line profiles correspond to a zero stimulation field, and donut-shaped line profiles require a large stimulation offset field. FFP, field-free-point; MPI, magnetic particle imaging.

TABLE 1 Symbols used in this study with their corresponding physical meanings.

Symbol	Physical meaning
H	Magnetic field
H_{AC}	AC excitation field amplitude
H_{DC}	DC field offset
G	Selection field gradient
T	AC excitation field period
ω_0	AC excitation field frequency
M_z	Magnetization along z direction
s_z	Signal along z direction
$S_{n,z}$	nth harmonic signal along z direction
s_{z_Debye}	Signal along z direction with Debye relaxation
S_{n,z_Debye}	nth harmonic signal along z direction with Debye relaxation
U_n	Chebyshev polynomials of the second kind
μ_0	Permeability of free space
m	Modulus of the magnetic moment of a single particle
k_B	Boltzmann constant
T^P	Particle temperature
β	Magnetic particle-related parameter
$\mathcal{L}$	Langevin function
M_0	Saturation magnetization
u	Heaviside step function
r_{Debye}	Debye relaxation kernel
$R_{n,Debye}$	nth relaxation frequency component
τ_B	Brownian relaxation time
η	Fluid dynamic viscosity
V_h	Hydrodynamic volume of magnetic nanoparticles
θ_{n,τ_B}	Delay phase angle due to relaxation effects

images, as visualized in Figure 9 and quantitatively indicated in Table 2. The STED-PSF-based images exhibited higher resolution and contrast than the Gaussian-PSF-based images. The resolution of the STED-PSF-based images was about 0.62 mm on the bar pattern phantom images.

The STED-PSF-based images showed weaker linearity between the concentration and signal intensity than the Gaussian-PSF-based images (Figure 10). The STED-PSF-based images exhibited less linearity with respect to the magnetic nanoparticle concentration, as shown in the bar concentration phantom image (Figure 10c), which may be caused by the negative signal generated during the subtraction of donut-PSF-based images from Gaussian-PSF-based images. There was non-uniformity in the neoplasm phantom (Figure 10c) and large background noise in the reconstructed phantom images (Figures 10c,d), which might be due to the low SNR of the donut PSF with a large DC offset.

5 | DISCUSSION

In this study, a donut-shaped PSF was generated in MPI by combining an excitation field with an offset stimulation field, which is similar to the excitation and stimulation laser beams in STED fluorescence microscopy.[42] The donut-shaped focal intensity distribution with zero intensity at the center in STED microscopy is created by a stimulated laser beam via a vortex phase mask. The center signal is depleted in STED-MPI because the center of the Langevin magnetization function is bypassed during the excitation period. High-resolution MPI images can be reconstructed by combining a small donut-shaped focal spot and a regular Gaussian focal spot, which was evaluated by theoretical deduction, imaging simulation, and scanner experiments in this study.

A small donut focal spot is essential for resolution improvement. The donut response is mainly affected by the DC offset field collinear to the AC excitation field and the receiver direction. We require a coaxial AC excitation and DC offset parallel to an FFL, such that the donut response is fixed on the plane perpendicular to the FFL. The donut PSF appears when the DC offset is greater than the AC excitation field. In this study, when the offset field is smaller than the excitation field amplitude, the signal profile of a point source is usually shown as a Gaussian or Lorentzian shape. A donut-shaped signal PSF can be formed when the offset field is greater than the excitation field. The donut radius gradually increased as the offset magnetic field increased. A donut PSF with a small radius can be selected to improve the resolution of MPI by subtracting it from a regular Gaussian-shaped PSF, which is applicable to FFL-based scanners with multi-view-angle projections.[43] In a non-FFP-based MPI scanner,[24] the donut signal has a relatively large radius and may have limited value in resolution enhancement.

The signal intensity of the donut focal spot is much smaller than that of the Gaussian focal spot with a zero offset field.[26] A higher excitation frequency can be used to improve the SNR, because the noise in the receiver electronics is dominated by a $1/\omega_0$ factor. The signal at the 3rd harmonic has a positive linear relationship with the excitation frequency.[44] Furthermore, the homogeneity improvement of the selection field[45] and signal-receiving chain optimization[46] may be used to increase the SNR of the donut-shaped focal spot.

The donut-shaped focal spot usually has a small SNR due to the large offset magnetic field, which may cause large background noise and affect the detection sensitivity of magnetic nanoparticles. When the offset is 50% of the excitation field, there may be another donut-shaped PSF in the 3rd harmonic signal map, which may correspond to the critical offset with a dramatic phase shift. The donut-shaped PSF exhibits a relatively larger radius and may not be applicable for STED image reconstruction. However, the donut-shaped focal spot with a strong signal may be used to improve the magnetic

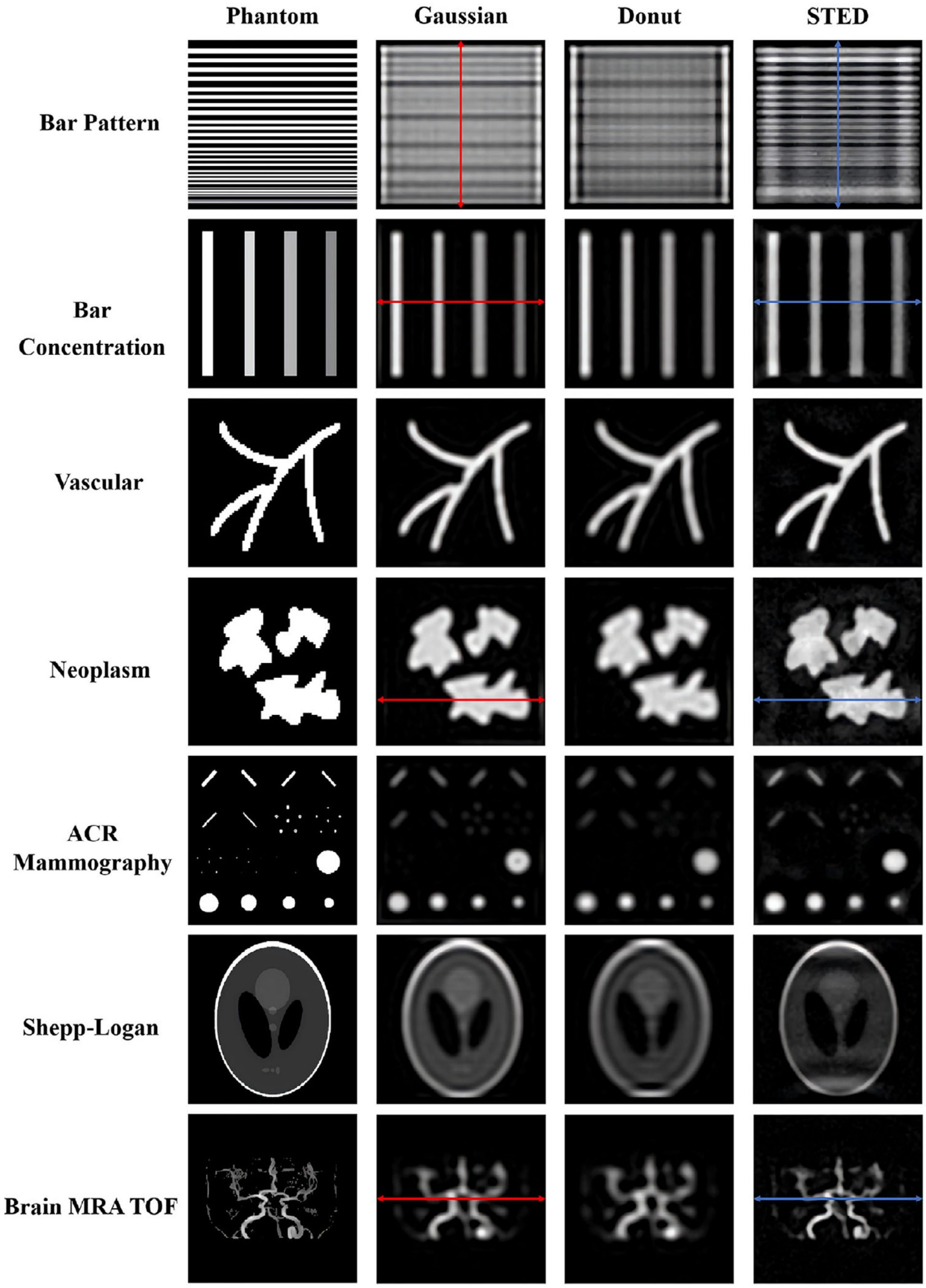

FIGURE 9 Reconstructed phantom images from Gaussian, donut, and STED kernels. The signal profiles of the blue and red lines are plotted in Figure 10. STED, stimulated emission depletion.

nanoparticle detection sensitivity by applying spatial pattern recognition algorithms,[47] which may be similar to MINSTED technology using a series of donut focal spots with different sizes.[22]

In STED-based image reconstruction, an STED factor was used to subtract the donut-based PSF or images and Gaussian-based PSF or images. The Gaussian-based PSF has a much higher SNR than the donut-based PSF. We multiplied the donut-based PSF by an STED factor to achieve a Gaussian-similar PSF after subtraction. If the STED factor is too large, the subtraction will generate negative signals and weaken the linearity between the signal intensity and magnetic particle concentration, which is similar to the "low image intensity" artifact in the immediate vicinity of the point source.[48] An optimized STED factor must be predefined based on the scanner and excitation parameter setting to achieve the best image quality. The iterative non-negative deconvolution algorithms can be used to facilitate the STED-MPI reconstruction process.[49]

TABLE 2 Reconstructed image quality assessment in STED-MPI phantom simulation.

Phantom	RMSE			PSNR			SSIM		
PSF style	Gaussian	Donut	STED	Gaussian	Donut	STED	Gaussian	Donut	STED
Bar pattern	10.22	10.26	7.78	64.36	64.30	71.18	0.033	0.019	0.192
Concentration bar	9. 80	9.08	5.50	65.24	66.84	76.68	0.177	0.215	0.710
Vascular	7.77	8.76	4.00	69.52	67.25	82.92	0.263	0.182	0.851
Neoplasm	9.91	9.31	4.90	67.18	66.15	79.21	0.256	0.220	0.695
ACR mammo	7.73	7.35	2.49	69.97	71.14	92.78	0.228	0.272	0.860
Shepp-Logan	8.85	8.94	5.97	67.26	67.05	74.91	0.296	0.302	0.523
Brain MRA TOF	6.27	6.58	3.16	74.07	72.94	88.33	0.398	0.363	0.825

Abbreviations: MPI, magnetic particle imaging; STED, STimulated emission depletion.

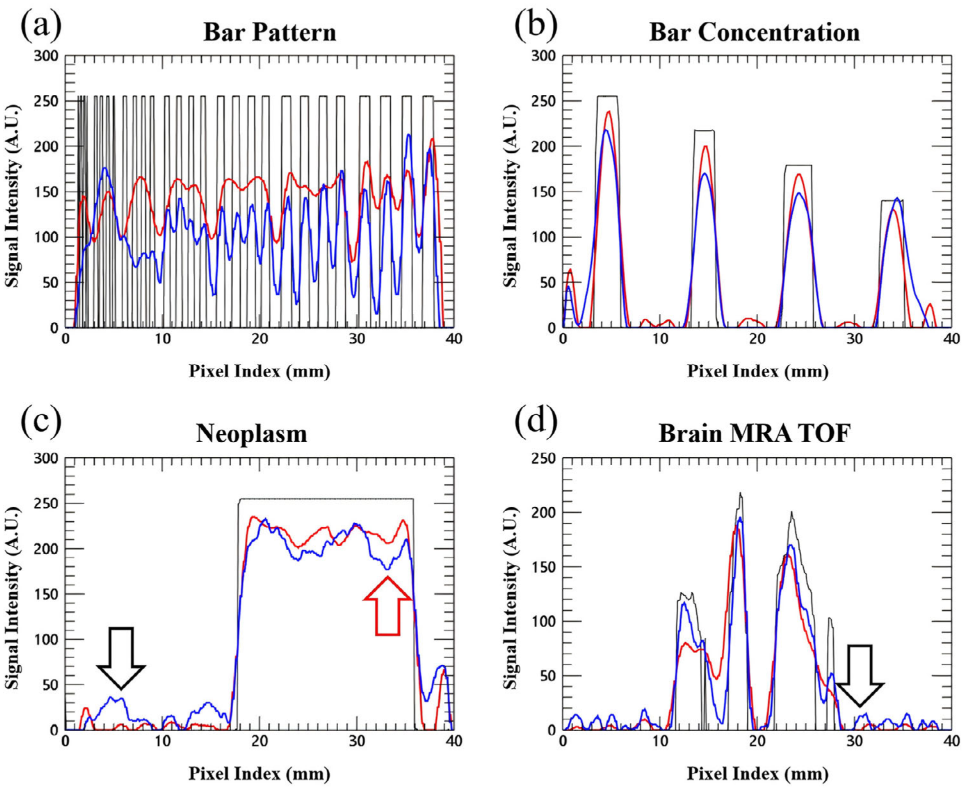

FIGURE 10 Line profiles from the reconstructed bar-pattern phantom image (a), bar-concentration phantom image (b), neoplasm phantom image (c), and brain MRA TOF phantom image (d). The red curves are the reconstructed line profiles based on the Gaussian PSF, the blue curve is based on the STED PSF, and black curves are the true phantom image line profiles. Red arrows indicate non-uniformity, and black arrows indicate background noise on the reconstructed images using STED-MPI. MPI, magnetic particle imaging; PSF, point spread function; STED, stimulated emission depletion.

The Debye relaxation model was used to mimic the relaxation effects on the harmonic signals in this study. The relaxation effects of the Langevin particles may induce a delay in magnetization and signal lag, which causes a phase change in the harmonics.[29] The Weizenecker Fokker–Planck formalism,[33] incorporating both Brownian and Néel relaxation, was developed to handle MPI simulations, which estimate many offset fields for each point in time.[32] The donut-shaped PSF exists in both the real and imaginary parts of the signal, which can be combined to improve the resolution using the STED imaging method. However, for larger particles, the relaxation time varies with the gradient offset field.[50] We prefer to use small particles for a homogeneous distribution of relaxation time, which is less dependent on the gradient selection fields.

The strength of the STED kernel is that it improves the resolution of MPI via a donut-shaped PSF with a smaller radius than the Gaussian kernel. The weakness of STED kernel is the low signal intensity due to the DC offset field, which is much greater than the magnitude of AC excitation field. The low signal intensity of the STED kernel resulted in non-uniformity and background noise in the reconstructed phantom images. Another weakness of STED-MPI is the long scan time and motion artifacts. STED-MPI requires two acquisitions, one with zero DC offset and the other with large DC offset, the subtraction of which may introduce motion artifact during living organism scan. Furthermore, the linearity of the FFL, homogeneity of the AC and DC fields, and uniformity of the gradient fields are all closely related to the STED imaging process and require calibration during image acquisition and reconstruction.[41] The available MPI scanner was an FFP-based scanner that showed the existence of donut-shaped line profiles in this study. We still need to build an FFL-based scanner with a switchable DC field to verify the STED imaging method on human or animal scans.

6 | CONCLUSION

This study proposed high-resolution MPI based on a small donut-shaped focal spot and the STED microscopic imaging principle. The donut focal spot can be generated by adding a large offset stimulation field parallel to the excitation and receiver directions. A focal spot modulation technique and deconvolution algorithm can be used to improve MPI image resolution.

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CONFLICT OF INTEREST STATEMENT

The authors have no conflicts to disclose.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable requests.

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